

Chi-Square Test

non-parametric hypothesis tests using the Chi-square statistic

Chi-Square

The χ^2 test (pronounced as chi-square test) is an important and popular test of hypothesis which fall is categorized in non-parametric test. This test was first introduced by Karl Pearson in the year 1900.

- ❑ It is used to find out whether there is any significant difference between observed frequencies and expected frequencies pertaining to any particular phenomenon.
- ❑ Here frequencies are shown in the different cells (categories) of a so-called contingency table. It is noteworthy that we take the observations in categorical form or rank order, but not in continuation or normal distribution.
- ❑ The test is applied to assess how likely the observed frequencies would be assuming the null hypothesis is true. This test is also useful in ascertaining the independence of two random variables based on observations of these variables.

This is a non parametric test which is being extensively used for the following reasons:

1. This test is a Distribution free method, which does not rely on assumptions that the data are drawn from a given parametric family of probability distributions.
2. This is easier to compute and simple enough to understand as compared to parametric test.
3. This test can be used in the situations where parametric test are not appropriate or measurements prohibit the use of parametric tests.

It is defined as:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where O refers to the observed frequencies and E refers to the expected frequencies.

Uses of Chi-Square Test

Chi Square test has a large number of applications where parametric tests can not be applied. Their uses can be summarized as under along with examples:

A test of independence.

1. This test is helpful in detecting the association between two or more attributes. Suppose we have N observations classified according to two attributes. By applying this test on the given observations (data) we try to find out whether the attributes have some association or they are independent.
2. This association may be positive, negative or absence of association. For example we can find out whether there is any association between regularity in class and division of passing of the students, similarly we can find out whether quinine is effective in controlling fever or not.
3. In order to test whether or not the attributes are associated we take the null hypothesis that there is no association in the attributes under study. In other words, the two attributes are independent.

4. After computing the value of chi square, we compare the calculated value with its corresponding critical value for the given degree of freedom at a certain level of significance.
5. If calculate value of 2 is less than critical or table value, null hypothesis is said to be accepted and it is concluded that two attributes have no association that means they are independent.
6. On the other hand, if the calculated value is greater than the table value, it means that the results of the experiment do not support the hypothesis and hypothesis is rejected, and it is concluded that the attributes are associated.

A total no. of 3759 individuals were interviewed in a public opinion survey on a political proposal. out of them 1872 were in the age group of above 50 years and the rest in the age group of 21 to ~~40~~ 50 years. A total of 2257 individuals were in favor of the proposal and 917 were opposed to it. A total of 243 person above 50 ; were undecided and 442 (21 to 50) were opposed to the proposal. Do you justify or contradict the hypothesis that there is no association between age group and attitude?

Solution:- A careful analysis of the problem results in the following table of contingency:-

		Favoured	opposed	undecided	Total
Age	above 50	1154	475	243	1872
	21 to 50	1103	442	342	1887
	Total	2257	917	585	3759

Ho: Age group and attitude are independent. i.e. there is
 no association between age group and attitude.

O	E (Rounded E)	O - E	(O - E) ²	$\frac{(O - E)^2}{E}$
1154	$\frac{1872 \times 2257}{3759} = 1124$	$1154 - 1124 = 30$	$(30)^2$	$30^2 / 1124 = 0.80$
475	$(1872 \times 917) / 3759 = 457$	18	18^2	$18^2 / 457 = 0.71$
243	$(1872 \times 585) / 3759 = 291$	48	48^2	$48^2 / 291 = 7.92$
1103	$(1887 \times 2257) / 3759 = 1133$	30 30	30^2	$30^2 / 1133 = 0.79$
442	$(1887 \times 917) / 3759 = 460$	18	18^2	$18^2 / 460 = 0.70$
342	$(1887 \times 585) / 3759 = 294$	48	48^2	$48^2 / 294 = 7.84$

$$E = \frac{\text{Row Total} \times \text{Column Total}}{N} = \frac{RT \times CT}{N}$$

$$\begin{aligned} \chi^2 &= \sum \sum \frac{(O - E)^2}{E} \\ &= 18.76 \\ v &= d.f. = (3-1)(2-1) \\ &= 2 \end{aligned}$$

$$\chi^2 \text{ calculated} \Rightarrow \chi^2_{\text{cal}} = 18.76$$

$$\text{For tabulated value d.f.} \Rightarrow (3-1)(2-1) = 2$$

For 95% level of significance

$$\chi^2_{0.05} (v=2) = 5.99$$

$$\text{Since } \chi^2 > \chi^2_{0.05}$$

H_0 is rejected.

Therefore age group and attitude are not independent.

$\therefore$ there is some association between age group and attitude.

Q. The following table gives for a sample of married women, the level of education and the marriage adjustment score!.

Level of Education	Marriage adjustment				Total
	very low	Low	High	very high	
College	24	97	62	58	241
High School	22	28	30	41	121
Middle School	32	10	11	20	73
Total	78	135	103	119	435

Can you conclude from the above data that the higher the level of education, the greater is the degree of adjustment in marriage?

Solution H_0 : there is no relation between the level of education and adjustment in marriage

$$v = (4-1)(3-1) = 6 ; \chi^2_{0.05}(v=6) = 12.59$$

O	E (rounded)	$(O-E)^2/E$
24	$43 = \frac{(241) \times (78)}{435}$	$19^2/43 = 8.40$
97	75	$22^2/75 = 6.45$
62	57	0.44
58	66	0.97
22	22	0.00
28	37	2.19
30	29	0.03
41	33	1.94
32	13	21.77
10	23	7.35
11	17	2.12
20	20	0.00

$$\chi^2 > \chi^2_{0.05}$$

$$\chi^2 = 57.66$$

$$\text{Since } \chi^2 > \chi^2_{0.05}$$

H_0 is rejected. That is, the level of education and adjustment in marriage are associated.

Thus we can conclude that the higher the level of education, the greater is the degree of adjustment in marriage.

Standard Chi-Square Distribution Table

T-10

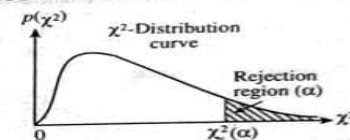
FUNDAMENTALS OF STATISTICS

TABLE VII : CRITICAL VALUES $\chi_{\nu}^2(\alpha)$ OF CHI-SQUARE DISTRIBUTION (RIGHT TAIL AREAS) FOR GIVEN PROBABILITY α ,

where

$$P[\chi^2 > \chi_{\nu}^2(\alpha)] = \alpha$$

AND ν IS DEGREES OF FREEDOM (df)



Degrees of Freedom (ν)	Probability (α)							
	.99	.975	.95	.90	.10	.05	.025	.01
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.634	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892

For large values of ν , the quantity $\sqrt{2\chi^2} - \sqrt{2\nu - 1}$ may be used as a standard normal variable.